

Disproof of Conjecture 00000005400: Orbit Distributions, Period Counts, and the Impossibility of a Conjugate Pair with Different Periods

TLMC submission — verdict: **FALSE**

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Abstract

Conjecture 00000005400 asserts two things at once: (i) there exist two maps with identical orbit distribution limits but different periodic point counts, and (ii) the separation is realized by an explicit *conjugate pair* with the same measure but different periods. We show that both layers fail in the literal discrete reading. First, on finite sets the periodic point count is a function of the orbit-length distribution, $P(f) = \sum_{l \geq 1} l N_l(f)$, so layer (i) is self-contradictory. Second, exact conjugacy preserves pointwise periods, $\text{per}_{\tau f \tau^{-1}}(\tau x) = \text{per}_f(x)$, so the witness objects demanded by layer (ii) — a conjugate pair with different periods — *do not exist*. We verify both identities by exhaustive machine enumeration: all 2304 triples $(\tau, f, x) \in S_4 \times S_4 \times \{0, 1, 2, 3\}$ for conjugacy, and all $4^4 = 256$ self-maps of a 4-point set for the counting identity. A Lean 4 development with zero axioms and zero `sorry` re-derives everything by kernel-checked computation.

1 The conjecture

Definition: Orbit distribution and period counting are two layers. *Conjecture:* There exist two maps with identical orbit distribution limits but different periodic point counts, and the separation is realized by an explicit conjugate pair with the same measure but different periods.

The conjecture treats “orbit distribution” and “period counting” as two independent layers that can come apart, and claims the coming-apart is exhibited by a conjugate pair. We refute both layers, in the natural discrete reading in which a *map* is a self-map of a set, the orbit distribution records cycle lengths, and conjugacy means $g = \tau f \tau^{-1}$ for a bijection τ .

2 Definitions

Let X be a set and $f: X \rightarrow X$.

Definition 1 (orbit, periodic point, period). For $x \in X$ write $f^0 x = x$ and $f^{k+1} x = f(f^k x)$. The point x is *periodic* if $f^k x = x$ for some $k \geq 1$; the least such k is the *period* $\text{per}_f(x)$. The *periodic orbit* of a periodic point x is $\mathcal{O}_f(x) = \{f^k x : k \geq 0\}$; all its elements are periodic with the same period, and $|\mathcal{O}_f(x)| = \text{per}_f(x)$.

Definition 2 (orbit distribution, periodic point count). For $l \geq 1$ let $N_l(f)$ be the number of distinct periodic orbits of f of length exactly l , and let $P(f)$ be the number of periodic points of f . The *orbit distribution* of f (its orbit profile) is the multiset $\{1^{N_1(f)}, 2^{N_2(f)}, 3^{N_3(f)}, \dots\}$, equivalently the sorted list of cycle lengths.

Definition 3 (conjugacy). Two maps $f, g: X \rightarrow X$ are *conjugate*, written $f \sim g$, if $g = \tau f \tau^{-1}$ for some bijection $\tau: X \rightarrow X$.

3 Layer 1 fails: the orbit distribution determines the periodic point count

Proposition 1 (Identity I). *For every self-map f of a finite set,*

$$P(f) = \sum_{l \geq 1} l N_l(f). \quad (1)$$

Consequently, if f and g have the same orbit distribution then $P(f) = P(g)$. There are no two maps with identical orbit distribution but different periodic point counts.

Proof. By Definition 1 the periodic points of f are partitioned into the distinct periodic orbits $\mathcal{O}$ of f , and each orbit of length l contains exactly l points, each of period l . Counting periodic points orbit by orbit, each orbit of length l is counted exactly once as an orbit in N_l and contributes exactly l points:

$$P(f) = \left| \bigsqcup_{\mathcal{O} \text{ orbit of } f} \mathcal{O} \right| = \sum_{\mathcal{O}} |\mathcal{O}| = \sum_{l \geq 1} \sum_{\substack{\mathcal{O} \\ |\mathcal{O}|=l}} l = \sum_{l \geq 1} l N_l(f).$$

The right-hand side is a function of the orbit distribution alone, so equal distributions force equal counts. $\square$

Identity I makes the first half of the conjecture self-contradictory: the two “layers” are not independent; the second is a function of the first.

Computational check. Over all $4^4 = 256$ self-maps of $X = \{0, 1, 2, 3\}$, the identity $P(f) = \sum l N_l(f)$ holds in all 256 cases, and among the 11 distinct orbit profiles no profile is realized with two different values of P (Table 1). The script `reproduce.py` (part B) re-runs this audit.

orbit profile	periodic count P	number of maps
(1)	1	64
(1, 1)	2	48
(1, 1, 1)	3	12
(1, 1, 1, 1)	4	1
(1, 1, 2)	4	6
(1, 2)	3	36
(1, 3)	4	8
(2)	2	48
(2, 2)	4	3
(3)	3	24
(4)	4	6

Table 1: All 256 self-maps of $\{0, 1, 2, 3\}$ grouped by orbit profile. Each of the 11 profiles carries exactly one periodic point count: the profile determines $P(f) = \sum l N_l(f)$.

4 Layer 2 fails: conjugacy preserves periods, so the demanded witnesses do not exist

Proposition 2 (Identity II). *Let $\tau: X \rightarrow X$ be a bijection and $g = \tau f \tau^{-1}$. Then for every $x \in X$ and every $k \geq 0$, $g^k(\tau x) = \tau(f^k x)$. In particular*

$$\text{per}_g(\tau x) = \text{per}_f(x) \quad \text{for all } x \in X. \quad (2)$$

So period is a conjugacy invariant, and no conjugate pair has different periods.

Proof. Induction on k . For $k = 0$ both sides equal τx . If $g^k(\tau x) = \tau(f^k x)$, then $g^{k+1}(\tau x) = g(g^k(\tau x)) = \tau f \tau^{-1}(\tau(f^k x)) = \tau(f(f^k x)) = \tau(f^{k+1} x)$, because $\tau^{-1} \tau = \text{id}$. Now $g^k(\tau x) = \tau x$ iff $\tau(f^k x) = \tau x$, and since τ is injective this holds iff $f^k x = x$. Taking the least such $k \geq 1$ on both sides gives $\text{per}_g(\tau x) = \text{per}_f(x)$. $\square$

Proposition 2 uses nothing about finiteness: for *any* two conjugate maps, the number of points of each exact period agrees. Therefore the second half of the conjecture — “the separation is realized by an explicit conjugate pair ... with different periods” — demands objects whose defining description is unsatisfiable. Whenever the first half could conceivably hold (non-conjugate maps sharing an orbit-distribution limit), the pair realizing it is, by that very clause, required to be conjugate — and conjugate pairs cannot separate. The conjunction is false under every reading.

Computational check. On $X = \{0, 1, 2, 3\}$ we enumerated all triples $(\tau, f, x) \in S_4 \times S_4 \times \{0, 1, 2, 3\}$: all 2304 satisfy $\text{per}_{\tau f \tau^{-1}}(\tau x) = \text{per}_f(x)$, zero violations. As an illustration, take $f = (01)(23)$, $\tau = (123)$, $g = \tau f \tau^{-1} = (02)(13)$; then every point of f and of g has period 2, and the pointwise comparison agrees on all four points (Table 2).

x	$\text{per}_f(x)$	$\tau(x)$	$\text{per}_{\tau f \tau^{-1}}(\tau x)$
0	2	0	2
1	2	2	2
2	2	3	2
3	2	1	2

Table 2: A concrete conjugate pair $f = (01)(23)$, $g = \tau f \tau^{-1}$ with $\tau = (123)$: pointwise periods are transported exactly.

For reference, the five cycle types of S_4 (partitions of 4) all give $P = 4$, as every point of a permutation is periodic:

cycle type (profile)	(1, 1, 1, 1)	(1, 1, 2)	(1, 3)	(2, 2)	(4)
number of permutations	1	6	8	3	6
periodic points P	4	4	4	4	4

Table 3: Cycle types of S_4 : 24 permutations, 5 profiles, one periodic count each.

5 Machine verification

The companion `lean4/` project (Lean 4.33.1, no Mathlib, no classical axioms) formalizes the 4-point case with permutation-valued and map-valued encodings:

- `conj_preserves_period` : Identity II, proved by `decide` over $24 \times 256 \times 4 = 24576$ configurations (τ over all permutations, f over *all* self-maps, x over all points);
- `count_from_profile_sum` : Identity I's bridge $P(f) = (\text{profile}(f)).\text{sum}$, decided over all 256 self-maps;
- `profile_det_count` : equal profiles force equal periodic counts, proved by rewriting with the bridge lemma;
- `permAtTable_eq` : kernel-checked certificate that the indexed maps `permAt` enumerate exactly the 24 permutations;
- `conjWitnessExists_false` and `refute` : the complete search for a conjugate pair with a pointwise period difference evaluates to `false`, and the conjecture's existential is negated by pure logic from `conj_preserves_period`.

`Check.lean` prints `does not depend on any axioms` for every theorem: the development is axiom-free in the strongest sense — not even `propext` or `Quot.sound` is used — and `sorry`-free, with all enumeration performed by the kernel.

6 Conclusion

- (a) The periodic point count is determined by the orbit distribution (Proposition 1); the conjecture's first layer cannot separate.
- (b) Period is a conjugacy invariant (Proposition 2); the conjecture's second layer demands conjugate witnesses with different periods, which do not exist.

The conjecture 00000005400 is therefore **false**: the alleged two-layer separation is internally inconsistent, and its realization by a conjugate pair is logically impossible.